WEEK 6

Set-Theoretic IR Models

Fuzzy Model and Extended Boolean Model



$A \cup B$ $A \cap B$ $\bar{A}$ $\bar{A} - B$

IR FOCUS

Implement flexible matching and fuzzy retrieval logic.

ADVANCED IR FOCUS

Membership functions, fuzzy scoring, and extended Boolean expressions.

PART 1 – LEARNING OUTCOMES

Learning Outcomes

By the end of Week 6, students will be able to:

1. **Explain** set-theoretic IR models.
2. **Understand** fuzzy relevance.
3. **Apply** membership values between 0 and 1.



4. **Explain** Extended Boolean Model.

A	B	A∩B	A∨B	$\bar{A}$
1	1	1	1	0
1	0	0	1	0
0	1	0	1	1
0	0	0	0	1

5. **Implement** flexible matching in an IR system.
6. **Mathematically** express fuzzy and extended Boolean retrieval.

$$\mu(d,q)=0.7$$

$$A \cap \bar{B} \vee C$$

7. **Connect** these models to practical IR applications and research opportunities.

What Are Set-Theoretic IR Models?

Set-theoretic IR models **extend classical Boolean search.**



They provide a mathematical framework based on **set theory** to represent and match information.

Classical Boolean search is strict:

Relevant

Not relevant

A document is either relevant or not relevant.

Set-theoretic models allow more flexible retrieval.



Degree of relevance

- Partial membership
- Gradual relevance
- Better handling of vagueness in queries and documents

They include:

1. **Fuzzy Model**
   Uses membership values between 0 and 1.

2. **Extended Boolean Model**
   Extends Boolean operators with additional power and flexibility.

Why Set-Theoretic Models Are Needed



Example

User query:

"*affordable smart health monitoring system*"



A document may not contain **all exact words**, but may still be useful and relevant.

Set-theoretic models help handle:

Partial relevance
Documents may match some, not all, of the query terms.

Flexible matching
Allows different ways to satisfy the user information need.

Ambiguous queries
Interprets queries that have multiple possible meanings.

User uncertainty
Models uncertainty in user intent and term preferences.

Soft inclusion and exclusion
Supports graded inclusion (not just in/out).

Fuzzy Model — Basic Idea

The **Fuzzy Model** treats relevance like a **dimmer switch**.

A document is not only relevant or irrelevant.
It can be:

- Highly relevant
- Moderately relevant
- Slightly relevant
- Not relevant



Relevance Levels

0.8 – 1.0	Highly relevant
0.5 – 0.8	Moderately relevant
0.2 – 0.5	Slightly relevant
0.0 – 0.2	Not relevant

Relevance is represented using values between **0 and 1**.

Example

Query: ***"smart health monitoring system"***

Doc 1
Smart health monitoring system for hospitals
Score: 0.92
(Highly relevant)

Doc 2
Health monitoring using IoT devices
Score: 0.68
(Moderately relevant)

Doc 3
Affordable wearable health tracker
Score: 0.41
(Slightly relevant)

Doc 4
Hospital management software
Score: 0.08
(Not relevant)

0.0 → 1.0
Relevance Score (0 = Not relevant, 1 = Highly relevant)

Fuzzy Membership Function

The membership function is:

$$\mu_A(x) \in [0, 1]$$

Where:

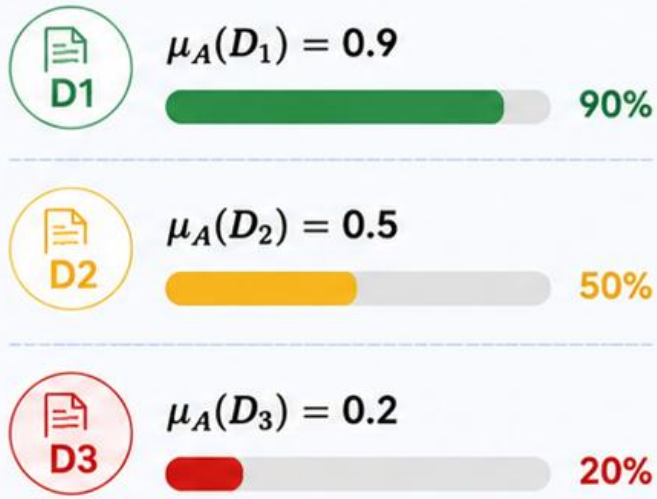
- A = set or concept
- x = document or term
- $\mu_A(x)$ = degree of membership or relevance

Membership on a Relevance Scale



Higher membership → Higher relevance

Example



Interpretation

Documents are **not simply relevant** or **irrelevant**.
Each document receives a **membership score between 0 and 1**, representing **how strongly** it belongs to the desired concept.



Fuzzy Model Example

Query:

"cute animals"



Different documents match the query to **different degrees**.

Documents and Fuzzy Membership Scores

Document	Description	Membership Value $\mu_A(x)$
D1	D1 talks a lot about cute animals.	0.9
D2	D2 talks a little about cute animals.	0.5
D3	D3 talks very little about cute animals.	0.2

0 Not relevant — 1 Highly relevant

Ranking (Higher is Better)

1. D1 — 0.9
2. D2 — 0.5
3. D3 — 0.2

Order of retrieval based on **fuzzy membership** values.

Computed Example of Fuzzy Retrieval

Query:

AI AND healthcare



We want documents that are about **AI** and **healthcare** to some degree.

For Document D1

D1

$\mu_{AI}(D_1) = 0.8$
(Relevance to "AI")

$\mu_{Healthcare}(D_1) = 0.6$
(Relevance to "healthcare")

Using Fuzzy AND

Fuzzy AND uses the minimum:

$$\mu(A \text{ AND } B) = \min(\mu_A, \mu_B)$$

$$\mu(D_1) = \min(0.8, 0.6)$$

$$\mu(D_1) = 0.6$$

Result

So, D1 has a fuzzy relevance score of **0.6**.

Interpretation:

D1 is somewhat relevant to both "AI" and "healthcare".
The fuzzy model captures partial relevance using values between 0 and 1.

0 — Not relevant
1 — Highly relevant

Extended Boolean Model – Basic Idea

The **Extended Boolean Model** improves normal Boolean search.

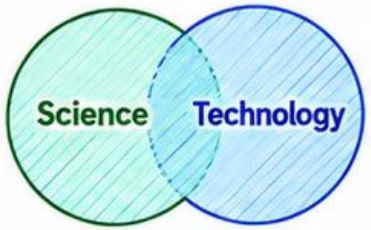
It lets users **include** and **exclude** terms more precisely.

Example Query

Science OR Technology BUT NOT Art

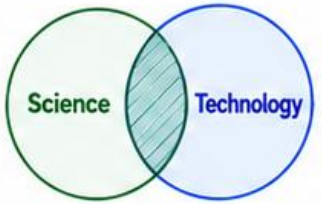
Meaning:
Retrieve documents about **science or technology** and **remove** documents about **art**.

How It Works (Set View)



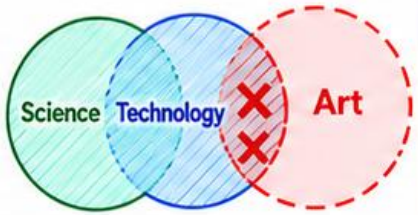
OR = Union
(Science ∪ Technology)

OR means documents in either set (or both).



AND = Intersection
(Science ∩ Technology)

AND means documents in both sets (common part only).



NOT Art = remove overlapping Art documents

Remove any documents that are also about art from the union (Science ∪ Technology).

Result

The system returns documents about **science** or **technology**, but **not** documents that **not** also about **art**.

OR widens the search.
NOT removes the unwanted part.

Key Takeaway: **OR = Union (∪)**, **AND = Intersection (∩)**, and **NOT = Exclusion / Difference (–)**.
Extended Boolean search combines inclusion and exclusion for more accurate and flexible retrieval.

Extended Boolean Model – Correct Set Representation

The Extended Boolean Model
improves normal Boolean search by allowing **inclusion (OR, AND)** and **exclusion (NOT)** more precisely.

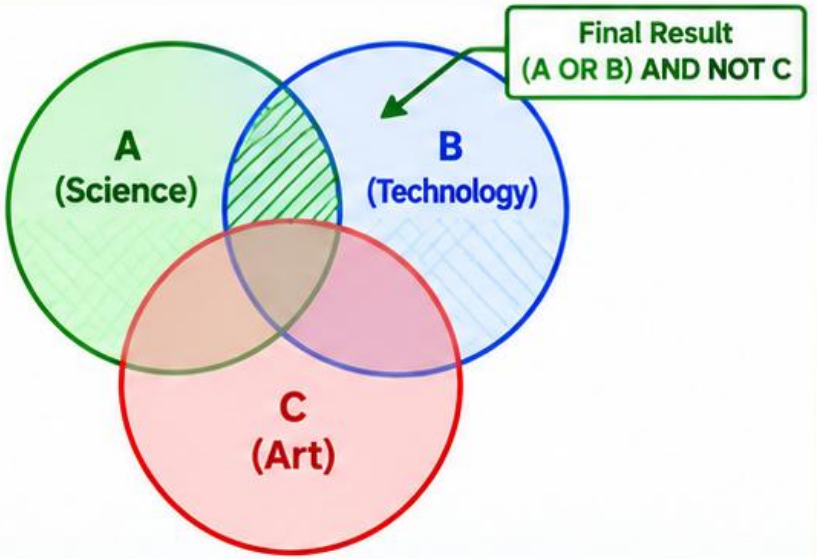
Sets (Example Domain)

A = Science
Set of documents related to science.

B = Technology
Set of documents related to technology.

C = Art
Set of documents related to art.

Venn Diagram (Set View)



Query (Mathematical Expression)

(A OR B) AND (NOT C)

Meaning

Retrieve documents related to **science or technology** (or both), but remove any documents that are related to **art**.

Legend

Included in final result (A OR B) AND NOT C

Excluded (any document in C)

Key Takeaway

Extended Boolean uses set operations:

- **OR** = **Union (∪)** → include documents from either set
- **AND** = **Intersection (∩)** → include documents from both sets
- **NOT** = **Difference (–)** → remove documents from a set

Simple Example

D1: Science and health	→ Included ✓
D2: Technology and AI	→ Included ✓
D3: Science and Art	→ Excluded ✗
D4: Art and culture	→ Excluded ✗
D5: Technology in education	→ Included ✓

Extended Boolean vs Fuzzy Model

Both models improve traditional Boolean retrieval, but they do it in **different ways.**



Feature	Fuzzy Model	Extended Boolean Model
Main Idea	Degree of relevance	Controlled Boolean expression
Output	Score between 0 and 1	More precise logical retrieval
Strength	Handles partial match	Handles include/exclude logic
Best Use	Flexible ranking	Expert search and filtering



IR Development Focus

For IR students, Week 6 must lead to a **flexible matching module**.



Required Features

1. Term relevance score
2. Partial matching
3. Fuzzy AND/OR/NOT
4. Extended Boolean query parser
5. Result ranking
6. Include/exclude filters
7. User-friendly search interface

What This Achieves

- More accurate retrieval
- Handles partial relevance
- Flexible and controllable search



Example



A medical document retrieval system can retrieve documents partially related to **symptoms**, **diseases**, and **treatments**.

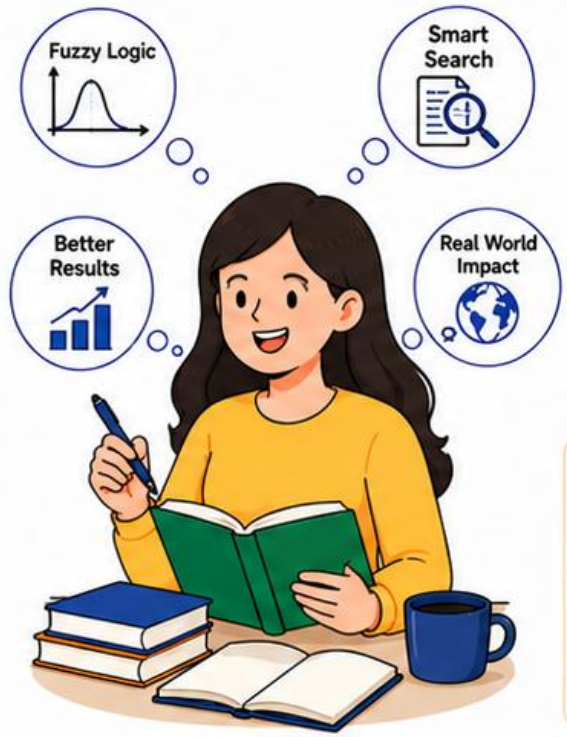
Advanced IR Research/Product Focus

For **Advanced IR students**, Week 6 must lead to **deep research** and **product innovation** in flexible retrieval.



Key Research & Product Outcomes

1. **Membership** function design
2. **Fuzzy** query scoring (0.75)
3. **Extended Boolean** mathematical expression (A OR B) AND (NOT C)
4. Comparison with **Boolean Model**
5. Research gap in flexible retrieval
6. **Commercial product idea** using fuzzy search
7. **Funding proposal** for smart search systems



Why It Matters

- Builds strong theoretical foundation and research capability
- Enables development of intelligent and user-centric IR systems
- Supports innovation, publications, and industry-ready products

Example



A **fuzzy search** system can return highly relevant medical documents even if the exact keywords are not present, **improving user satisfaction and decision making.**

Case Study Assignment

For IR Students



Implement a **fuzzy search module** for a selected FYP.



Minimum Requirements

1. User query input
2. Term matching
3. Relevance score between 0 and 1
4. Fuzzy AND/OR/NOT
5. Extended Boolean query support
6. Ranked results

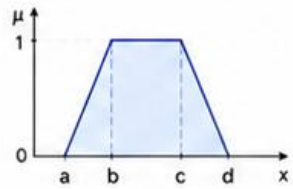


For Advanced IR Students

Write a short **research/product** note on **fuzzy retrieval** with membership function, formula, computed example, and possible commercial application.



1. Membership Function



2. Fuzzy Scoring Formula

Fuzzy AND:

$\mu(A \text{ AND } B) = \min(\mu A, \mu B)$

Fuzzy OR:

$\mu(A \text{ OR } B) = \max(\mu A, \mu B)$

Fuzzy NOT:

$\mu(\text{NOT } A) = 1 - \mu A$

3. Computed Example

Query: **"diabetes AND treatment"**

Document	μ(diabetes)	μ(treatment)	AND Score
D1	0.85	0.60	0.60
D2	0.70	0.90	0.70
D3	0.40	0.80	0.40

Ranking: D2 (0.70) > D1 (0.60) > D3 (0.40)

4. Extended Boolean Example

(diabetes OR hypertension) AND (NOT surgery)



5. Possible Commercial Application

Medical Search · Legal Discovery · Enterprise Search · E-commerce Search · and more

Focus on **accuracy**, **flexibility**, and **real-world impact**.

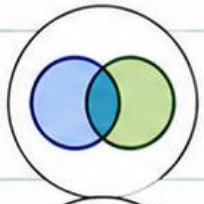
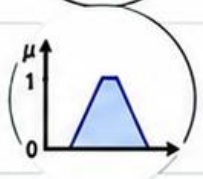
Quiz

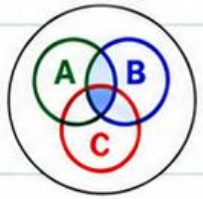
1. What is the difference between classical Boolean and fuzzy retrieval?
2. What does $\mu A(x)$ mean?
3. What is the range of fuzzy membership?
4. Calculate fuzzy AND for 0.7 and 0.4.
5. What does (A OR B) AND (NOT C) mean?
6. How can fuzzy search improve an FYP product?



Summary

Week 6 covered:

1.  Set-theoretic IR models
2.  Fuzzy relevance
3. 

Membership function
4. 

Fuzzy AND/OR/NOT
5. 

Extended Boolean Model
6.  Flexible matching
7. 

FYP implementation
8.  Research/product/funding direction

